

Question 1 (23 Marks) Start this question on a new page. **Marks**

- a) Calculate correct to 2 decimal places **1**

$$5^{\frac{3}{2}} \times 8^{\frac{4}{5}} \times 3^2$$

- b) Evaluate and express your answer using scientific notation **2**

$$\frac{3 \times 10^7 - 2 \times 10^6}{4 \times 10^3 + 2 \times 10^3}$$

- c) Simplify

(i) $3xy^2 \times -4x^2$ **1**

(ii) $\frac{a^3b^7}{a^2(b^3)^3}$ **1**

- d) Evaluate $|-2| + |-5|$ **1**

- e) Express $0.\dot{0}4$ as a simplified fraction **1**

- f) Harry and Sally paid \$315.00 for a meal at a restaurant. This included a $12\frac{1}{2}\%$ tip. What was the cost of the meal without the tip? **2**

- g) Expand and simplify completely

(i) $3x(x-7) - 4x(x-7)$ **2**

(ii) $4(2x+1)^2$ **2**

- h) Factorise completely

(i) $27x^3 - 8y^3$ **1**

(ii) $3x^2 + 16x + 20$ **1**

(iii) $(6x^2 - 6) + (12x^2 - 36x - 48)$ **3**

Question 1 continues on Page 2

Question 1 continued**Marks**

i) Simplify as a single fraction in its simplest form:

$$(i) \quad \frac{5x+3}{4} - \frac{2x-3}{5} \quad 2$$

$$(ii) \quad \frac{x^2 - 14x + 13}{(x+2)(x+3)} \times \frac{x^2 + 6x + 9}{(x-1)^2} \quad 3$$

Question 2 (21 marks) Start this question on a new page.a) Simplify completely 2

$$\sqrt{48} - \sqrt{44} + \sqrt{75}$$

b) Simplify with a rational denominator 2

$$\frac{\sqrt{5} + 2}{\sqrt{5} - 3}$$

c) Solve $2x^2 - 8x + 5 = 0$ leaving your answer as an exact value in its simplest form. 2d) Solve the following simultaneous equations: 3
$$\begin{aligned} 6x + 3y &= 0 \\ 7x - 4y &= -5 \end{aligned}$$
e) Find x if $4^{3-x} = 8^x$ 2f) If $A = P(1+r)^n$, find r if: $A = 3345$, $P = 2500$ and $n = 4$ 2
Answer correct to 2 decimal places*Question 2 continues on Page 3*

Question 2 continued**Marks**

g) Solve the following:

$$|2x - 5| = 7$$

2

h) Solve and graph on the number line

(i) $2 \leq 4(x - 2) < 8$

3

(ii) $|3x - 2| \leq 5$

3**Question 3** (28 marks) Start this question on a new page.

a) Sketch each of the following curves on a separate diagram.

10

Clearly show all important features, including x and y intercepts, asymptotes, maximum and minimum values where necessary.

(i) $y = x^2 + 3$

(ii) $x^2 + y^2 = 25$

(iii) $y = 3^x$

(iv) $y = |x - 4|$

(v) $y = \frac{5}{x - 3}$

b) State the domain and range of

4

(i) $y = \sqrt{4 - x^2}$

(ii) $y = 4^x$

c) Using the graphs from part (a) determine whether the following graphs are functions or relations:

(i) $y = x^2 + 3$

1

(ii) $x^2 + y^2 = 25$

1*Question 3 continues on Page 4*

Question 3 continued**Marks**

d) Sketch the following piecemeal function:

4

$$f(x) = \begin{cases} 3 & x \geq 5 \\ -x & 0 \leq x < 5 \\ x+5 & x < 0 \end{cases}$$

e) Determine whether the function $f(x) = x^4 - 3x^2 + 2$ is odd, even or neither.

2

f) Given $f(x) = x^3 + 7x + 5$, find $f(-1)$

1

g) Given $g(x) = x^2 - 6$ find values for x for which $g(x) = 3$

1

h) Shade the region on the number plane where $y \geq x^2 - 2$ and $y > 6 + 3x$ hold simultaneously. Indicate any points of intersection.

4**END OF PAPER**

SOLUTIONS

Question 1	(23 Marks)	<u>Start this question on a new page.</u>	Marks
a)	531.09		1
b)	$\frac{14000}{3}$ $= 4.667 \times 10^3$	1 mark for each or 2 for correct final answer	2
c)	(i) $-12x^3y^2$		1
	(ii) $\frac{a}{b^2}$		1
d)	7		1
e)	$\frac{2}{45}$		1
f)	315=112.5% 1 mark 2.8=1% 280= 100% cost without tip is \$280		2
g)	Expand and simplify completely		
	(i) $3x^2 - 21x - 4x^2 + 28x$ $= -x^2 + 7x$ or $7x - x^2$	1 mark correct expansion	2
	(ii) $4(4x^2 + 4x + 1)$ $16x^2 + 16x + 4$	1 mark correct expansion	2
h)	Factorise completely		
	(i) $(3x - 2y)(9x^2 + 6xy + 4y^2)$		1
	(ii) $(x + 2)(3x + 10)$		1

SOLUTIONS

$$\begin{aligned}
 & 6x^2 - 6 + 12x^2 - 36x - 48 && 1 \text{ mark expansion} \\
 & = 18x^2 - 36x - 54 \\
 & = 18(x^2 - 2x - 3) && 1 \text{ mark HCF} \\
 & = 18(x-3)(x+1) && 1 \text{ mark factorising}
 \end{aligned}$$

3

Question 1 continued

$$\begin{aligned}
 & \frac{5x+3}{4} - \frac{2x-3}{5} \\
 & \frac{5(5x+3) - 4(2x-3)}{20} \\
 \text{i)} & = \frac{25x+15-8x+12}{20} && 2 \\
 & = \frac{17x+27}{20} && 1 \text{ mark correct expansion} \\
 & && 1 \text{ mark}
 \end{aligned}$$

$$\begin{aligned}
 \text{(i)} & \frac{(x-13)(x-1)}{(x+2)(x+3)} \times \frac{(x+3)^2}{(x-1)^2} && 2 \text{ mark factorising} \\
 & = \frac{(x-13)(x+3)}{(x+2)(x-1)} && 1 \text{ mark simplifying}
 \end{aligned}$$

3

Question 2 (21 marks) Start this question on a new page.

a) 2

$$\begin{aligned}
 & 4\sqrt{3} - 2\sqrt{11} + 5\sqrt{3} && 1 \text{ mark simplifying} \\
 & = 8\sqrt{3} - 2\sqrt{11} && 1 \text{ mark collecting like terms}
 \end{aligned}$$

b) 2

$$\begin{aligned}
 & \frac{\sqrt{5}+2}{\sqrt{5}-3} \times \frac{\sqrt{5}+3}{\sqrt{5}+3} \\
 & = \frac{5+3\sqrt{5}+2\sqrt{5}+6}{5-9} && 1 \text{ mark} \\
 & = \frac{11+5\sqrt{5}}{-4} && 1 \text{ mark}
 \end{aligned}$$

c)

2

$$2x^2 - 8x + 5 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{8 \pm \sqrt{8^2 - 4 \cdot 2 \cdot 5}}{2 \cdot 2}$$

1 mark

$$= \frac{8 \pm \sqrt{24}}{4}$$

$$= \frac{8 \pm 2\sqrt{6}}{4}$$

$$= \frac{4 \pm \sqrt{6}}{2}$$

1 mark

d)

$$6x + 3y = 0 \dots\dots\dots (1) \times 4$$

$$7x - 4y = -5 \dots\dots\dots (2) \times 3$$

$$24x + 12y = 0 \dots\dots\dots (3)$$

$$21x - 12y = -15 \dots\dots\dots (4)$$

1 mark

$$45x \dots\dots = -15 \dots\dots (3 + 4)$$

$$x = \frac{-1}{3}$$

subst with eqn 1

$$6\frac{1}{3} + 3y = 0$$

$$3y = -2$$

$$y = \frac{-2}{3}$$

$$x = \frac{1}{3} \text{ and } y = \frac{-2}{3}$$

1 mark each

$$2^{2(3-x)} = 2^{3x}$$

$$6 - 2x = 3x$$

1 mark

$$e) \quad 6 = 5x$$

2

$$x = \frac{6}{5}$$

1 mark

$$3345 = 2500(1+r)^4 \quad 1 \text{ subst}$$

$$1.338 = (1+r)^4$$

$$f) \sqrt[4]{1.338} = 1+r$$

$$1.075508949 = 1+r$$

$$r = \cancel{0.75508949} \quad 0.075508949$$

$$r = 0.76 \text{ (2d.p)} \quad 1 \text{ mark}$$

$$0.08.$$

2

$$g) \quad |2x-5| = 7$$

2

$$2x-5 = 7$$

$$2x = 12$$

$$x = 6$$

$$-(2x-5) = 7$$

$$-2x+5 = 7$$

$$-2x = 2$$

$$x = -1$$

$$\text{Check: } |2 \times 6 - 5| = 7 \text{ ok} \quad |2 \times -1 - 5| = 7 \quad 1 \text{ mark each}$$

h) Solve and graph on the number line

$$(i) \quad 2 \leq 4(x-2) < 8$$

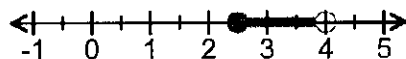
3

$$2 \leq 4x-8 < 8$$

$$10 \leq 4x < 16$$

1 mark each value

$$\frac{5}{2} \leq x < 4$$



1 mark correct graph of solutions given

(ii)

3

$$3x - 2 \leq 5$$

$$3x \leq 7$$

$$x \leq \frac{7}{3}$$

$$-3x + 2 \leq 5$$

$$\text{or } -3x \leq 3$$

$$x \geq -1$$

Check values: $3\left(\frac{7}{3}\right) - 2 \leq 5$ ok $3(-1) - 2 \leq 5$ ok 1 mark each



1 mark correct graph for solution $-1 \leq x \leq \frac{7}{3}$

Question 3 (28 marks) Start this question on a new page.

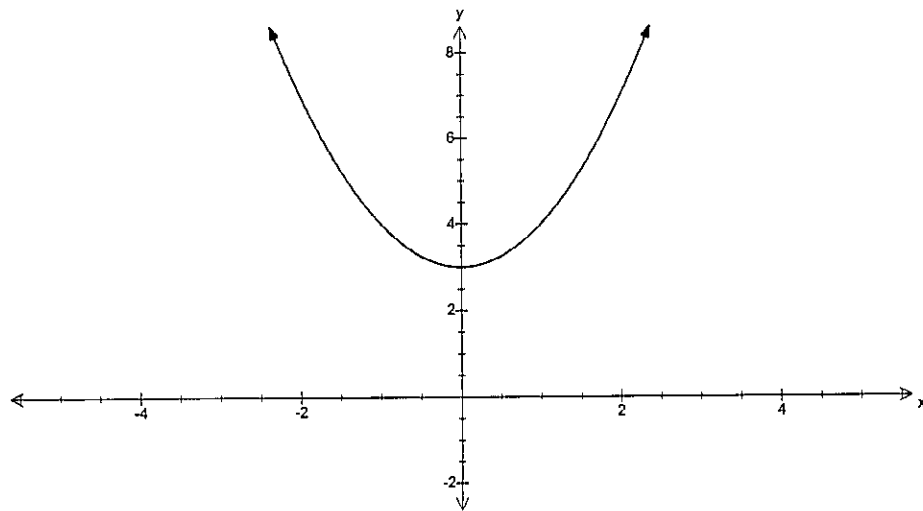
a) 2 marks for each graph

10

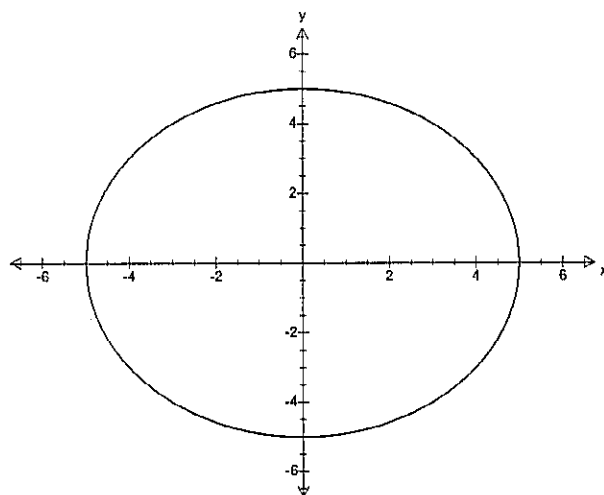
Clearly show all important features, including x and y intercepts, asymptotes, maximum and minimum values where necessary.

(i) $y = x^2 + 3$

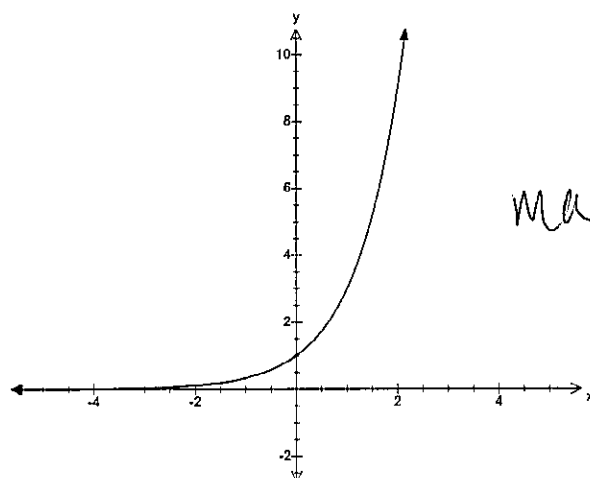
max turning point at $(0,3) = y$ int.



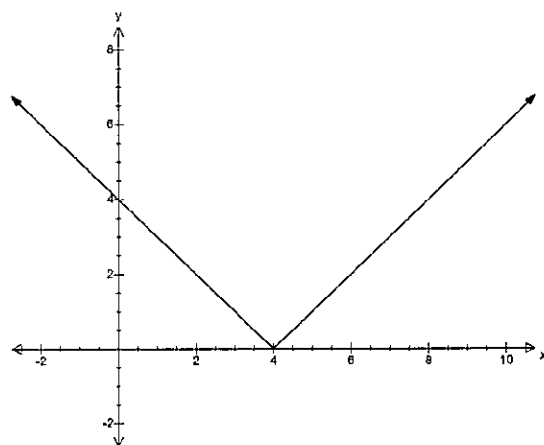
(ii) $x^2 + y^2 = 25$

 x int $(-5,0)$ and $(0,-5)$ y int $(0,-5)$ and $(0,5)$ 

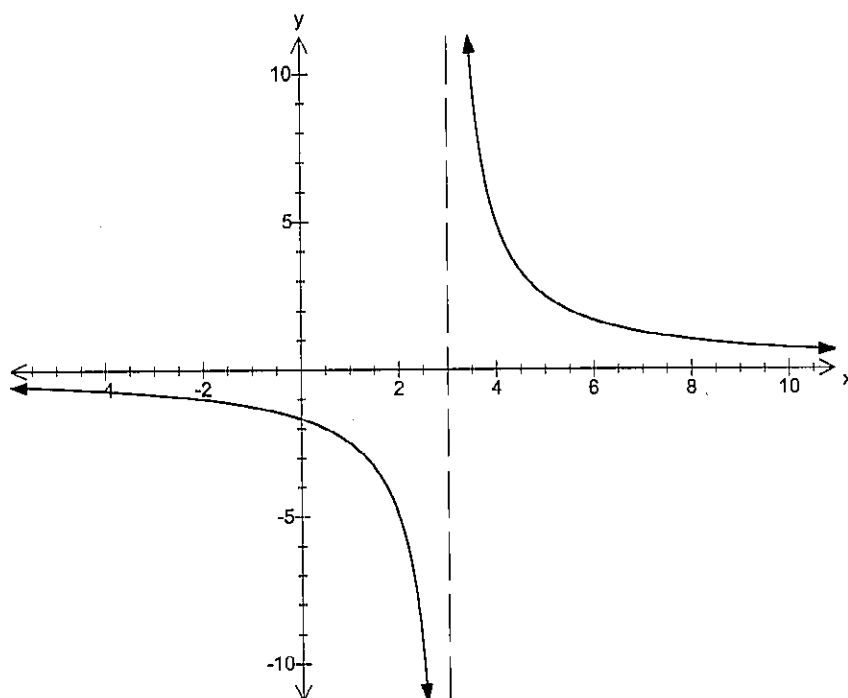
(iii) $y = 3^x$

 y int $(0,1)$ *mark one other point*

(iv) $y = |x - 4|$ x int $(4,0)$ and y int $(0,4)$



(v) $y = \frac{5}{x-3}$

vert ass ($x=3$)

b) State the domain and range of

4

(i) Domain: $-2 \leq x \leq 2$ Range: $0 \leq y \leq 2$ 1 mark each(ii) Domain: all real x Range: ~~all~~ $y > 0$ 1 mark each

c) Using the graphs from part (a) determine whether the following graphs are functions or relations:

(i) function

1

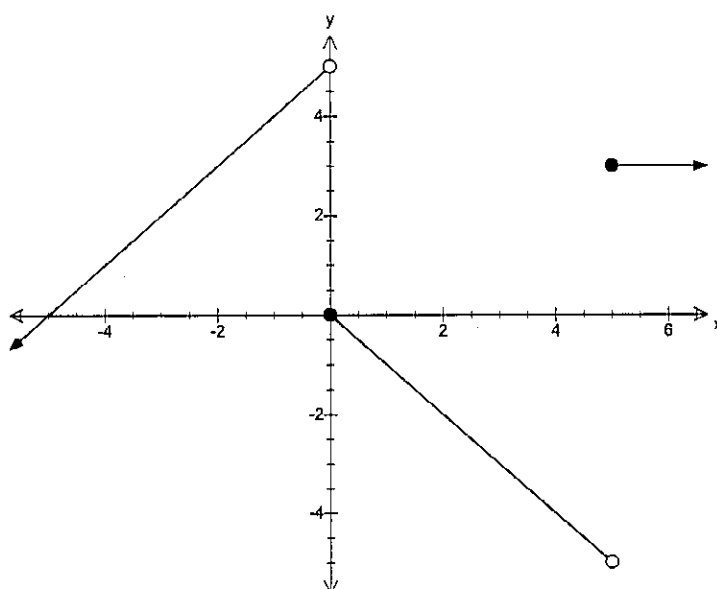
(ii) relation

1

d) Sketch the following piecemeal function:

4

$$f(x) = \begin{cases} 3 & x \geq 5 \\ -x & 0 \leq x < 5 \\ x+5 & x < 0 \end{cases} \quad \text{1 for each graph and 1 for correct limits}$$



e)

2

$$f(-x) = (-x)^4 - 3(-x)^2 + 2$$

$$f(-x) = x^4 - 3x^2 + 2 \quad \text{1 mark}$$

$$f(x) = f(-x) \quad \text{Even} \quad \text{1 mark}$$

$$\begin{aligned} \text{f) } f(-1) &= (-1)^3 + 7(-1) + 5 \\ f(-1) &= -3 \end{aligned}$$

1

$$g(x) = 3 = x^2 - 6$$

$$\text{g) } x^2 = 9$$

$$x = \pm 3$$

1

- h) Shade the region on the number plane where
- $y \geq x^2 - 2$
- and
- $y > 6 + 3x$

4

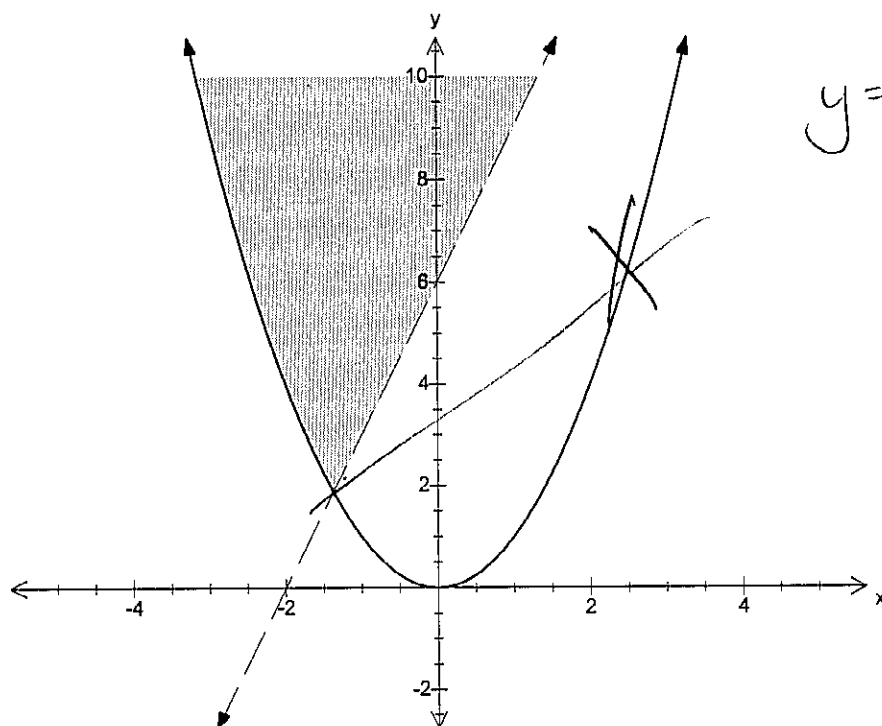
points of intersection

$x=1.701$

$y=0.8953$

1 for each graph

1 for shading

~~1 for point of intersection~~*1 dotted/solid line* *$y = x^2 - 2$
needs to be
typed in.*